Deep Metric Learning-Based Feature Embedding for Hyperspectral Image Classification

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***Abstract*— Learning from a limited number of labeled samples (pixels) remains a key challenge in the hyperspectral image (HSI) classification. To address this issue, we propose a deep metric learning-based feature embedding model, which can meet the tasks both for same- and cross-scene HSI classifications. In the first task, when only a few labeled samples are available, we employ ideas from metric learning based on deep embedding features and make a similarity learning between pairs of samples. In this case, the proposed model can learn well to compare whether two samples belong to the same class. In another task, when an HSI image (target scene) that needs to be classified is not labeled at all, the embedding model can learn from another similar HSI image (source scene) with sufficient labeled samples and then transfer to the target model by using an unsupervised domain adaptation technique, which not only employs the adversarial approach to make the embedding features from the source and target samples indistinguishable but also encourages the target scene's embeddings to form similar clusters with the source scene one. After the domain adaptation between the HSIs of the two scenes is finished, any traditional HSI classifier can be used. In a simple manner, the nearest neighbor (NN) algorithm is selected as the classifier for the classification tasks throughout this article. The experimental results from a series of popular HSIs demonstrate the advantages of the proposed model both in the same- and cross-scene classification tasks.**

***Index Terms*— Cross-scene classification, deep metric learning, domain adaptation, hyperspectral image (HSI), small sample set (3S) classification.**

I. Introduction

HYPERSPECTRAL imaging is one of the most important techniques in remote sensing (RS), which can provide a large volume of hyperspectral images (HSIs) that can be used for a wide variety of applications related to earth observation, such as precision agriculture and environmental monitoring [1]. HSIs captured by hyperspectral imaging sensors usually consist of several hundred spectral bands from the same area on the surface of the Earth. This abundant spectral information makes the accurate discrimination of different land-cover classes possible. HSI classification, therefore, has been one of the most popular fields in research of the hyperspectral community [2]. The purpose of HSI classification is to predict a unique label to each pixel so that it can be well represented by a given land-cover class [3]. For that, numerous supervised classification algorithms, such as support vector machines (SVMs) [4], random forest [5], extreme learning machines [6], and neural networks [7], have been heavily studied over the past few decades. Most of these conventional supervised classifiers can be seen in [1]. Although the HSI classification has achieved a huge breakthrough, it still has challenges. One of the main reasons is that the number of labeled samples (pixels) is usually limited in practice (caused by the difficulty of manual labeling). Therefore, how to achieve satisfying classification performances with a small number of training samples has attracted many researchers in recent years.

To partially address this small sample set (3S) issue, active learning and semisupervised learning have been widely studied. Active learning, whose purpose is to find the most informative samples for training, is an interactive information-extraction approach [8]–[10]. Active learning begins with the training by a small initial labeled set and then selects a few additional samples from the unlabeled samples through an iterative process [11]. By selecting the most useful samples for training, the model can be improved with the same number of training samples, and thus, the problem of limited training samples can be alleviated. In contrast, semisupervised approaches, such as semisupervised SVMs [12], [13], graph-based semisupervised learning [14], and manifold-based semisupervised learning [15], directly utilize the unlabeled data to enhance the generalization ability of the model. More recently, the manifold semisupervised regularization is proposed for exploiting the local structure preserving [16], [17]. However, these existing approaches either not focus exactly on the HSI classification or still should require a great number of training samples (e.g., more than 200 training samples per class) for good generalization.

Incorporating the spatial information into the HSI classification is considered as a powerful strategy to tackle the

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problem of lacking labeled samples [18]. This is because homogeneous regions in an HSI are generally composed by many interconnected pixels, and therefore, neighboring pixels in the same HSI are usually related or correlated [2]. For this reason, a large number of classification algorithms based on spectral–spatial information have been developed, such as morphological profile-based methods [19]–[23], Gabor wavelet-based methods [24], [25], and superpixel-based methods [26], [27]. However, most of these methods heavily rely on handcrafted features and thus often have a lot of parameters that need to be set manually and these hyperparameters are usually data sensitive. Therefore, it is difficult for them to reach an optimal balance between discriminability and robustness.

Recently, deep learning-based methods have drawn increasing attention in the field of the HSI classification [28]. The first attempt in this direction can be found in [29], where stacked autoencoders are used to extract useful high-level features. Subsequently, an increasing number of methods using deep learning techniques have been proposed, such as 1-D convolutional neural network (CNN) [30], 2-D CNN [31], [32], 3-D CNN [33], [34], and recurrent neural network (RNN) [35]. Although these deep learning-based methods have achieved great success in HSI classification tasks, they all need a large amount of labeled samples to train their large number of parameters. To better deal with such problem, many strategies have been used. For example, by constructing pixel-pair samples, the number of training samples is significantly increased [36]. At the same time, Yu *et al.* [37] have proposed a novel CNN model by using data augmentation, dropout functions, and 1×1 convolutional kernels, which has got a huge progress in the field of deep learning-based HSI classification. More recently, in [38] and [39], semisupervised CNNs have also been designed to improve the classification performance by using a smaller number of training data. Based on the advantages of PCANet [40], several advanced networks, such as R-VCANet [41] and MugNet [42], are proposed for 3S HSI classification. In this article, we would like to construct a deep learning network that can reach the comparative results using much less training samples (e.g., only three training samples per class).

Another possible way to address the problem of lacking labeled samples is to transfer knowledge from another similar data. Actually, in the real applications in HSI classification, we usually meet a situation; the target image that needs to be classified has not available labeled samples [43]. For example, it is impossible to classify an HSI in real time if we have to label the pixels manually in the same image before implementing classification. In this case, a natural and reasonable idea is to make use of data information from another similar source image (including sufficient labeled samples) to help the target image classification. This task is thus called cross-scene classification defined in [43]. Caused by various imaging factors (e.g., atmosphere or light conditions), there is significant spectral shift between different scene images, which means that the pixels from source and target HSI scenes may vary in spectral distribution even though they belong to the same land-cover class. Therefore, all the aforementioned approaches are no longer valid for this kind of classification.

In order to reduce the spectral shift between the cross-scene images, applying domain adaptation techniques is the most popular choice. In the field of the HSI classification, the common algorithms for domain adaptation can roughly be categorized into two classes: classifier-level domain adaptation [44]–[46] and feature-level domain adaptation [43], [47]. The first class makes the classifier generalize to the target scenes' domain by tuning the parameters and structures of the model during the training procedure. For the second class, the features in source and target scenes will be mapped into a common domain space where the distributions of the two data sets are matched. Recently, Ye *et al.* [43] have proposed a feature-level domain adaptation method based on dictionary learning. In this approach, the spectral features from both the source and target scene images share with the same atoms in the dictionary so that the extracted features from both the scenes are in the shared subspace that is represented by these atoms. This makes a promising performance in the cross-scene HSI classification. However, learning a common dictionary is difficult because the optimization procedure is complex.

In this article, we propose a novel deep metric learning-based framework, which can meet the tasks both for the same-scene (training and testing samples in the same HSI) and cross-scene classifications. The goal of the proposed framework is to project the input features into a well-defined metric space (e.g., Euclidean space) where the mapping features have small within-class spacing and large between-class spacing. The novelty of our framework is that we present a powerful deep embedding model for HSI features' extraction by using metric learning-based loss functions so that the extracting features can be used for any classifier based on this specific metric space. More specifically, for the task of the supervised classification, we define a deep metric space instead of Euclidean space to further improve the resulting performance, in which the model can learn well to compare whether two samples belong to the same class even though only a few labeled samples are used for training. On the other hand, when there is no labeled sample on the target scene image, our model can first learn from the source scene images and then transfer to the target model by applying unsupervised domain adaptation techniques. Once the domain adaptation is learned, the source and target data can be embedded into a shared space so that the model training from the source scene can be used for the target scene classification. Compared with other related state-of-the-art approaches, our method has three main contributions, which are summarized as follows.

1) First, we propose a deep metric learning-based model for HSI classification. Based on such model, we present a well-defined network architecture, which shows promising classification results with limited training samples.
2) Second, we develop a novel deep metric learning-based training algorithm for the proposed model, which can learn the similarity between a pair of samples (pixels). Such training strategy makes the training process much

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stable and achieves a significant improvement in classification results comparing with the previous methods.

3) Finally, the proposed deep leaning framework can apply both for the same- and cross-scene HSI classifications; to the best of our knowledge, it is the first attempt to apply deep learning for cross-scene HSI classification. We hope that this article opens the door to more deep learning-based cross-scene HSI classification in the future.

When we apply our method to the supervised classification for same-scene HSIs, our method can outperform the previous state-of-the-art deep learning-based approaches with only a few number of training samples. We also apply our method to the cross-scene HSI classification and confirm that our model achieves superior or comparable performance in comparison with the previous works.

The rest of this article is organized as follows. Section II introduces some recent works related to our method, including metric learning-based and domain adaptation-based techniques. In Section III, the proposed supervised model and unsupervised domain adaptation methods are presented. The experiments on both the same- and cross-scene classifications are reported in Section IV. Finally, in Section V, we provide some conclusions.

II. Related Work

Our approach is largely related to recent research on the fields of deep metric learning for recognition and unsupervised domain adaptation with adversarial training networks.

A. Deep Metric Learning

The goal of the metric learning is to learn a distance metric such that samples belonging to the same classes are close as possible in a self-defined metric space and samples from different classes are as possible to be far from each other. For example, Yu *et al.* [48] utilized pairwise constraints to integrate the semantic similarity in distance metric learning. According to Weinberger and Saul [49], a number of researchers have demonstrated that learning an appropriate distance metric from labeled samples can lead to significant improvements in many visual classification tasks. By introducing deep neural networks for feature embedding, deep metric learning-based methods have been then increasingly proposed [50]–[53]. Among these works, triplet networks [54] as a deep metric learning model achieved a successful application for classification learning with only few training data. After then, relation networks [55] were proposed to learn a comparing model that is able to transfer to measure the relation scores between the images, being a powerful few-shot learning model. However, all these approaches are designed for specific learning tasks (e.g., RGB images classification), which are not valid for applying directly to the pixel-level HSI classification. To achieve this task, we design a neural network architecture specifically, which is novel and powerful, and comparing to the other networks designed based on conventional CNNs, such as the networks presented by Li *et al.* [33], Hamida *et al.* [34], and Yu *et al.* [37], our network shows a significant improvement in the results of 3S classification. To the best of our knowledge, we are the first to apply metric learning that is based on the triplet loss as well as the mean square error (mse) loss for HSI classification.

B. Unsupervised Domain Adaptation

The topic of unsupervised domain adaptation also shows an increasing interest in recent years, which allows us to learn a model from a labeled source data sets and then well perform on a different target data sets where the labeled data are nonexistent. Recently, the main strategy of the unsupervised domain adaptation is to guide feature learning by minimizing the discrepancy between the source and target feature distributions [56]–[58]. For this purpose, adversarial training techniques are commonly used. More specifically, by designing different optimizations which related to the discriminative learning task on source domain and the domain-invariance task with respect the shift between two domains, various adversarial training-based methods have been presented. For instance, Ganin and Lempitsky [57] added a gradient reversal layer directly between the feature extractor and the domain classifier so that the unsupervised domain adaptation can be achieved during a standard backpropagation training, while in [56], the hidden representations of all the task-specific layers are projected to a reproducing kernel Hilbert space to reduce the domain discrepancy.

The most related methodologies to ours are the adversarial discriminative domain adaptation (ADDA) of [59] and metric-based ADDA (M-ADDA) of [60]. These approaches focus on learning an effective embedding of target samples to the source feature space by confusing a domain discriminator. They all use the pretrained source model as an initialization for the target model and then turn the target model parameters only during adversarial training. However, comparing to the ADDA that uses a standard supervised loss for source data classification, M-ADDA applies triplet loss to cluster the source data, showing a better classification performance on digits' adaptation data sets of MNIST and USPS. Inspired by these works, we design a novel network architecture in this article, which is the first try to meet the task for cross-scene HSI classification by using metric learning-based domain adaptation techniques. Although we use the same loss function as M-ADDA method for domain adaptation, there are two main differences that we need to clarify. First, we apply domain adaptation loss directly to the embedding features instead of the encoder features in M-ADDA (in M-ADDA, an embedding network module include an encoder network and a decoder network, and the encoder feature is produced by the encoder network). In this way, all the parameters in the target embedding module can be updated during the process of domain adaptation. The second difference is that our network architecture designed for HSI classification is substantially different from the structure of M-ADDA which is set for natural images' (e.g., RGB images) labeling. Our task is a pixel-level classification, and we exploit all convolution layers with 1×1 kernels so that the number of parameters can be significantly reduced to satisfy our learning process by using a 3S.

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TABLE I
NOTATIONS USED IN THIS ARTICLE

Notations	Descriptions
$\mathcal{X}$	training sample set (labeled pixels in HSIs)
$\mathcal{S}$	sample set of source HSI data
$\mathcal{T}$	sample set of target HSI data
P	patch size of each sample
N	number of spectral bands
K	number of classes
E_φ	embedding module parameterized by φ
M_ϕ	metric module parameterized by ϕ
D_θ	discriminator parameterized by θ
$m_{i,j}$	metric value between samples x_i and x_j

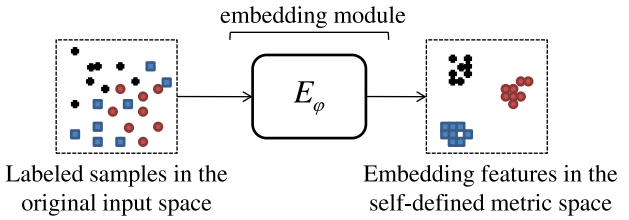


Fig. 1. Embedding module is a feature extractor, which is trained to make the embedding features to form clusters for each category.

III. PROPOSED METHOD

In order to address the challenge of lacking labeled samples in HSI classification, first, we build a deep metric learning-based model to embed the samples to a specific metric space where the similarity or distance between any two samples is clearly represented. This model can learn well with only a few labeled samples. Once the distance metric is learned, we adopt a nearest neighbor (NN) classifier to simply implement the classification tasks (considering the small number of training samples). Furthermore, when the classified images (target images) have no labeled samples, an unsupervised domain adaptation technique is presented so that the deep metric learning-based model trained on another similar scene images (source images that have sufficient labeled samples) can also be applied to the target scene images.

To begin with, we introduce some important notations that are used in this article, as listed in Table I.

A. Deep Metric Learning-Based Model

In the proposed deep metric learning-based model, the main goal is to learn a deep embedding module E_φ (parameterized by φ) such that it can well project the input samples into a self-defined metric space (the embedding feature space) where the samples with the same labels are pulled together and those with different labels are pushed away from each other. For example, when there are three classes, Fig. 1 shows the purpose of embedding model.

In order to learn the embedding module, we have to give a specific definition to the embedding feature space, in which we can accurately measure what is close and far between the samples. We assume that there are K classes in the training set $\mathcal{X} = \{(x_i, y_i)\}_i^n$ with n training samples in total and each sample x_i has a label value y_i. As shown in Fig. 2, for any two

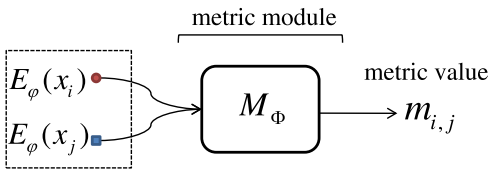


Fig. 2. Given two samples x_i and x_j, their metric value is measured by two steps. First, they both be fed into the embedding module to generate the corresponding embedding features $E_\varphi(x_i)$ and $E_\varphi(x_j)$, and then, the embedding features are concatenated together into the metric module to output the metric value (a distance or similarity value).

samples x_i and x_j, after being embedded into the self-defined metric space, their metric value can be calculated through the metric module M_ϕ. In the following, we present two different definitions of the metric module M_ϕ and their optimization algorithms.

1) Distance-Based Euclidean Metric Module: When the self-defined metric space is a Euclidean space, the metric value between two samples is a distance metric that is defined as follows:

$$m_{i,j} = \|E_\varphi(x_i) - E_\varphi(x_j)\|^2. \tag{1}$$

In this case, the parameter set ϕ of the metric module M_ϕ is empty. Therefore, we only need to train the parameter set φ, which can be learned by minimizing the triplet loss

$$\varphi \leftarrow \arg\min_{\varphi} \sum_{(x_p, x_q, x_l)} \max(m_{p,q} - m_{p,l} + 1, 0) \tag{2}$$

where samples x_p and x_q have the same label and x_p and x_l have a different label. It is clear that (2) encourages the embedding of x_p to be closer to x_q than to x_l by at least margin 1. Optimizing such term results in moving the samples with the same label into a cluster and pushing those with different labels far away from each other. Therefore, an expected embedding shown in Fig. 1 can be formed as much as possible, which allows us to efficiently implement classification by using the NN classifier based on the metric distance that we defined in (1) (Euclidean distance would be in this case).

2) Similarity-Based Deep Metric Module: Instead of directly using Euclidean distance to measure the metric value between two embedding features, in this module, we apply deep learning technique to further learn their similarity. In other words, the metric module M_ϕ is a neural network, which produces a metric value representing the similarity between any two embeddings

$$m_{i,j} = M_\phi(\mathcal{C}(E_\varphi(x_i), E_\varphi(x_j))) \tag{3}$$

where $\mathcal{C}(.,.)$ is an operator to be concatenation of feature embeddings in depth. In a simple manner, we make this output metric value $m_{i,j}$ in the range from 0 to 1 to indicate the degree of similarity between x_i and x_j. The higher the metric value, the greater the similarity.

In this case, an mse loss function will be used to train the model, because we consider the metric value prediction as a

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Algorithm 1 Training the S-DMM in a Single Iteration. RANDOMSAMPLE($\mathcal{X}, N$) Denotes a Set of N Elements Chosen Uniformly at Random From Set $\mathcal{X}$, Without Replacement

Input: Model (E_φ, M_ϕ), and training set $\mathcal{X} = \{(x_1, y_1), ..., (x_n, y_n)\}$, where each $y_i \in \{1, ..., K\}$. $\mathcal{X}_k$ denotes the subset of $\mathcal{X}$ containing all elements (x_i, y_i) such that $y_i = k$.
Output: Updated model (E_φ, M_ϕ).
begin
  $\mathcal{A} \leftarrow \emptyset$; $\mathcal{B} \leftarrow \emptyset$;
  for i *in* $\{1, ..., K\}$ **do**
    $\mathcal{A} \leftarrow \mathcal{A} \cup$ RANDOMSAMPLE$(\mathcal{X}_i, 1)$;
    $\mathcal{B} \leftarrow \mathcal{B} \cup$ RANDOMSAMPLE$(\mathcal{X}_i, \lceil |\mathcal{X}_i|/2 \rceil)$;
  end
  $L \leftarrow 0$;
  for (x, y) *in* $\mathcal{A}$ **do**
    for (x', y') *in* $\mathcal{B}$ **do**
      $m_{i,j} \leftarrow M_\phi(\mathcal{C}(E_\varphi(x), E_\varphi(x')))$;
      $L \leftarrow L + (m_{i,j} - \mathbf{1}(y == y'))^2$;
    end
  end
  Update parameter sets φ and ϕ by backpropagating through L.
end

regression problem, which is formulated as

$$\varphi, \phi \leftarrow \arg\min_{\varphi,\phi} \sum_{x_i, x_j \in \mathcal{X}} (m_{i,j} - \mathbf{1}(y_i == y_j))^2 \tag{4}$$

where the sample pairs have similarity 1 if they have the same label and those with different labels have similarity 0. By optimizing (4), the model can learn to measure how much the similarity that two samples have and to compare whether they belong to the same class. Algorithm 1 shows the procedures for training this model on the labeled sample set $\mathcal{X}$. Once trained, it can be used for classification by applying the NN classifier based on the similarity that we defined in (3). In this case, the NN of a sample is the one who has the biggest similarity with it.

B. Unsupervised Domain Adaptation

When the labeled samples in the HSI images that we need to classify are not existent, we design a powerful network architecture to meet the task for cross-scene HSI classification. To achieve this goal, we first train an embedding module E_{φ_S} on source scene images by optimizing (2) and then fix the source embedding module E_{φ_S} and train the target embedding module E_{φ_T} by applying an unsupervised domain adaptation technique. Considering that a source data set $\mathcal{S}$ and a target data set $\mathcal{T}$ are drawn from two different distributions, the purpose of such technique is to a learn embedding function on the target data so that it can map the target data into the target embeddings that have the same distribution as the source embeddings. Fig. 3 shows an example of the processes in

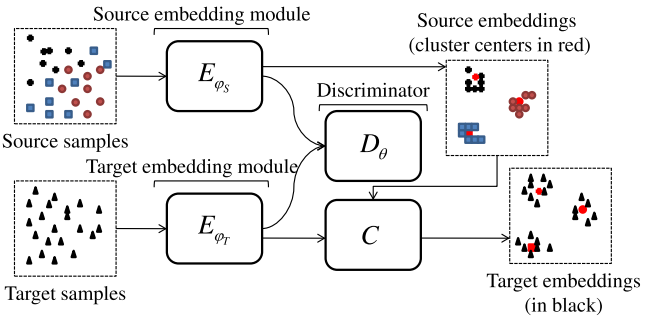


Fig. 3. Overview of the unsupervised domain adaptation model. The source embedding module is trained first on source labeled data and then fixes the source embedding module and updates the target embedding module by optimizing a discriminator [by minimizing (6)] and a center magnet loss [by minimizing (7)] together.

training the target embedding module, which is achieved by optimizing the following two loss terms:

$$\mathcal{L}(\varphi_T, \theta) = \mathcal{L}_A(\varphi_T, \theta) + \mathcal{L}_C(\varphi_T) \tag{5}$$

where φ_T is the parameter set for the target embedding module E_{φ_T} and θ corresponds to the parameters of a discriminator module D_θ that we use to adapt the distributions between the source and target feature embeddings, $E_{\varphi_S}(\mathcal{S})$ and $E_{\varphi_T}(\mathcal{T})$. We achieve this purpose by optimizing

$$\mathcal{L}_A(\varphi_T, \theta) = \min_\theta \max_{\varphi_T} - \sum_{s_i \in \mathcal{S}} \log D_\theta(E_{\varphi_S}(s_i)) - \sum_{t_i \in \mathcal{T}} \log(1 - D_\theta(E_{\varphi_T}(t_i))). \tag{6}$$

More specifically, the parameter set θ is trained to make the discriminator D_θ to distinguish the embeddings' mapping from source data or target data, while parameter set φ_T is trained to confuse the discriminator at the same time. By jointly optimizing the discriminator and target embedding function, two embedding features $E_{\varphi_S}(\mathcal{S})$ and $E_{\varphi_T}(\mathcal{T})$ can be encouraged to be distinguishable in their distributions.

Apart from using an adversarial learning approach to maximizing the domain confusion discussed earlier, we achieve domain adaptation also by minimizing the center magnet loss that is defined as

$$\mathcal{L}_C(\varphi_T) = \sum_{t_i \in \mathcal{T}} \min_k \|E_{\varphi_T}(t_i) - C_k\|^2 \tag{7}$$

where C is the set of cluster centers corresponding to the source embeddings, represented as red dots in Fig. 3. Each center C_k is calculated by the mean of the embedding features belonging to the class k. Minimizing the center magnet loss encourages the target embeddings to form similar clusters as the source embeddings. This is useful because in this way, the corresponding distributions of the source and target embeddings are obviously in the same space, which can help to achieve the domain confusion as well.

Once the unsupervised domain adaptation model is trained enough, any classifier learned from source embeddings can be used directly for target embeddings classification. In this article, we simply use the NN classifier for the task. As we

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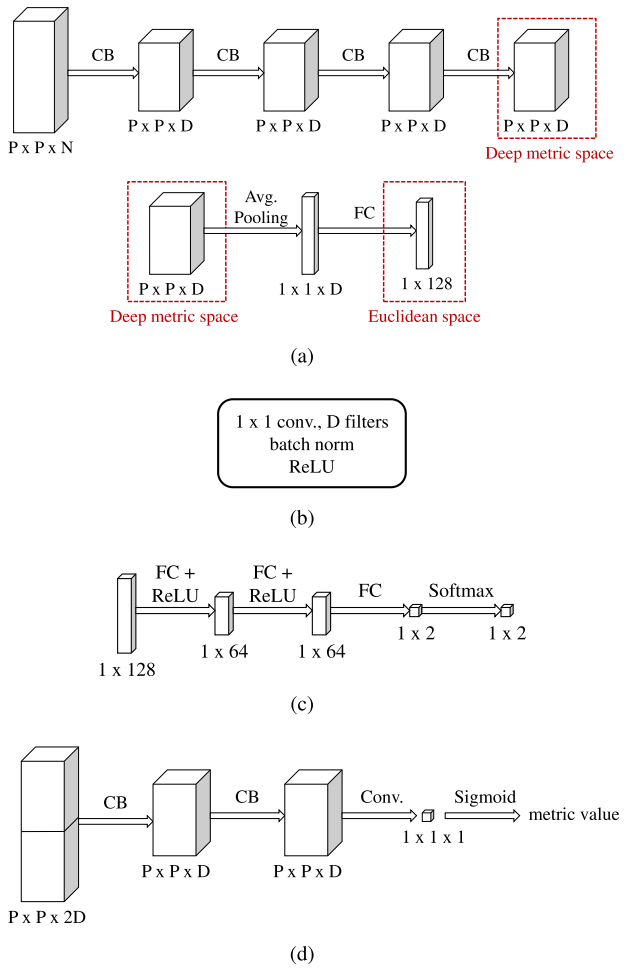


Fig. 4. Deep metric learning-based network architecture for HSI classification. (a) Embedding module architecture. (b) CB. (c) Discriminator module architecture. (d) Metric module architecture.

train our model mainly by applying the M-ADDA algorithm [60], we point the reader to [60] for more details about this component.

C. Network Architecture

In the same-scene HSI classification, a few number of labeled pixels are selected as a training sample set, and the goal of the task is to classify each pixel vector that is in the same-scene image as the training pixels to a specific given category (e.g., woods, corn, grass, and so on), while in the cross-scene HSI classification, the labeled pixels are selected in the source scene image, and the task is to classify pixels in the target scene image. For such tasks, we design an efficient and simple network architecture for each module what mentioned earlier, as shown in Fig. 4. In order to comprehensively utilize the spectral and spatial information, each pixel and its neighbors (with a window size of $P \times P$) in the same HSI are taken as a sample input of the network, and therefore, the size of each sample is $P \times P \times N$, where N is the number of channels (spectral bands) in HSIs.

As shown in Fig. 4(a), there are four convolutional blocks (CBs) in our embedding module E_φ. It is worth noting that for embedding features to the Euclidean space, an average pooling and a fully connection layer should be added. The CB, shown in Fig. 4(b), contains a 1×1 convolution with D number of filters, a batch normalization, and a rectified linear unit (ReLU) nonlinearity activation function. Since we use 1×1 convolution kernel, the number of parameters in the network is significantly reduced, which is helpful when only a few number of labeled samples are used for training. The discriminator module, whose objective is to predict whether inputs are from source or target data, is constructed as shown in Fig. 4(c), where softmax is used as the final activation function. Fig. 4(d) shows the architecture of the metric module, where two CBs are first employed, followed by a $P \times P$ convolution layer, and the final output layer is the sigmoid function in order to generate a metric value in a range between 0 and 1.

IV. Experiments

To illustrate the effectiveness of the proposed method for HSI classification, including the same- and cross-scene classification tasks, we apply it to five popular real-world data sets comparing with various state of the arts. The performance metrics that we use in this article are overall accuracy (OA), average accuracy (AA), as well as kappa coefficient (κ). The OA score evaluates the overall classification accuracy, which is the number of the correctly classified samples (pixels) of all classes divided by the total size of testing samples and the AA score is the average of each classification accuracy per class. κ, on the other hand, is a statistical measure related to the degree of agreement for categorical items [61]. Clearly, higher values of these three metrics indicate a better classification performance. Empirically, the patch size of a pixel sample, i.e., P, is fixed to 5 for all data sets, and the numbers of the 1×1 filters in the CB, i.e., D, are set to 64 for the same-scene classification and 200 for the cross-scene classification. Since these two parameters are fixed, our method does not have any other parameters related to the models that need to be set manually. As for training settings, we use PyTorch software [62] to run our training procedures with the stochastic gradient descent (SGD) optimization algorithm. The learning process will be stopped after 50 000 training iterations for the proposed model without using the validation set, and the learning rate of to 0.1, divided by 10 for the later half of iterations. All the source codes related to the model and training settings are available at https://github.com/szubing/S-DMM.

In order to better describe in the following, we name our supervised models as Euclidean distance-based deep metric model (ED-DMM) and similarity-based deep metric model (S-DMM), and name our unsupervised model as ED-DMM with unsupervised domain adaptation (ED-DMM-UDA), which is for cross-scene HSI classification.

A. Same-Scene Classification

In the same-scene classification, the training and testing samples (pixels) are from the same-scene HSIs, which can be considered as a general situation in the HSI classification. In this task, three real data sets, including Pavia University scene (PaviaU), Salinas scene (Salinas), and Kennedy

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TABLE II

NAME AND SAMPLES' SIZE OF EACH LAND-COVER CLASS IN THE PAVIAU DATA SET

Class	Name	Samples Size
$C1$	Asphalt	6631
$C2$	Meadow	18649
$C3$	Gravel	2099
$C4$	Trees	3064
$C5$	Metal sheets	1345
$C6$	Bare Soil	5029
$C7$	Bitumen	1330
$C8$	Bricks	3682
$C9$	Shadows	947
	Total	42776

TABLE III

NAME AND SAMPLES SIZE OF EACH LAND-COVER CLASS IN THE SALINAS DATA SET

Class	Name	Samples Size
$C1$	Brocoli green weeds 1	2009
$C2$	Brocoli green weeds 2	3726
$C3$	Fallow	1976
$C4$	Fallow rough plow	1394
$C5$	Fallow smooth	2678
$C6$	Stubble	3959
$C7$	Celery	3579
$C8$	Grapes untrained	11271
$C9$	Soil vinyard develop	6203
$C10$	Corn senesced green weeds	3278
$C11$	Lettuce romaine 4wk	1068
$C12$	Lettuce romaine 5wk	1927
$C13$	Lettuce romaine 6wk	916
$C14$	Lettuce romaine 7wk	1070
$C15$	Vinyard untrained	7268
$C16$	Vinyard vertical trellis	1807
	Total	54129

Space Center (KSC),[1] are used in the experiments, which are described as follows

1) The PaviaU data set was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor showing the area in Pavia, Northern Italy. After removing 12 noisy bands from the original 115 spectral bands, 103 bands are kept for processing. The spatial size of the image is 610×340 with the spatial resolution of 1.3 m, and it covers 9 classes with 42 776 labeled pixels in total, as shown in Table II. The ground-truth map of the PaviaU scene image can be seen in Fig. 10(b).
2) The Salinas data set was acquired by another sensor called Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), which was imaged over the area of the Salinas Valley, CA, USA. It has a spatial resolution of 3.7 m and a size of $512 \times 217 \times 204$ (after omitting the noisy bands). The Salinas data set has 54 129 labeled pixels in total with 16 land-cover classes, and the detail information is shown in Table III and Fig. 5.
3) The KSC data set was acquired also by the AVIRIS instrument when it was flying over the Kennedy Space Center, FL, USA, but has a lower spatial resolution of 18 m. The original spatial dimension of these data is 512×614, and there are 176 spectral bands for analysis. As shown in Table IV, there are 5211 labeled pixels in total with 13 classes representing the various land-cover materials. Fig. 6 shows the false-color image and ground-truth map of the KSC data.

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

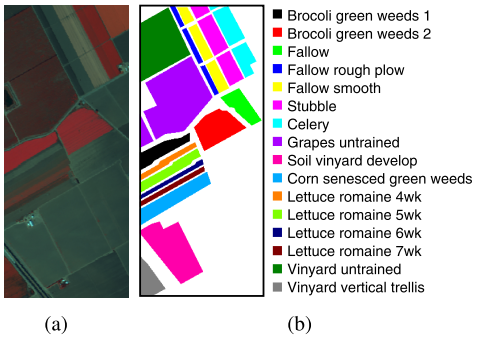


Fig. 5. Salinas data set. (a) False-color image. (b) Ground-truth map.

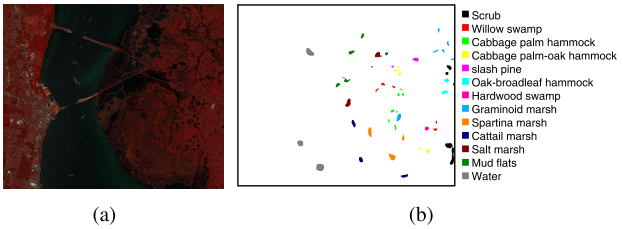


Fig. 6. KSC data set. (a) False-color image. (b) Ground-truth map.

The most advanced deep learning classification algorithms proposed recently are compared with our ED-DMM and S-DMM methods. These approaches include the 3-D CNN model [34], spectral–spatial CNN (SS-CNN) model [33], and 2-D CNN model [37]. Since our model selects the NN as a classifier for the extracted features, we also provide such classifier that directly applies to the original spectral features as the baseline. Since we are interested in the scenarios of 3S, a very few number of labeled samples per class (e.g., 3, 5, 10, 20, 30, 40, and 50) are randomly picked out from the labeled set to constitute the training set, while the rest of the labeled samples are used for testing. Each experiment is carried out ten times to decrease the sampling effect. For a fair comparison, it should be noted that we did not apply data augmentation to further improve the classification performance for all methods except for the 2-D CNN model (because the data augmentation is the main technique in the 2-D CNN model), although only a few number of training samples are used.

Figs. 7–9 show the classification performances, including OA, AA, as well as the kappa coefficient (κ) of all comparing methods in three data sets with different numbers of training samples per class. Here, each experiment is repeated randomly ten times, and both the mean and the standard variation are reported. From Figs. 7–9, we can clearly see that our S-DMM model always outperforms the others with the highest classification accuracy and the best robustness. Although both NN and our ED-DMM methods use the same NN classifier based on the Euclidean distance, the features extracted by the ED-DMM model are more representative than the original spectral features. Our S-DMM model also uses NN

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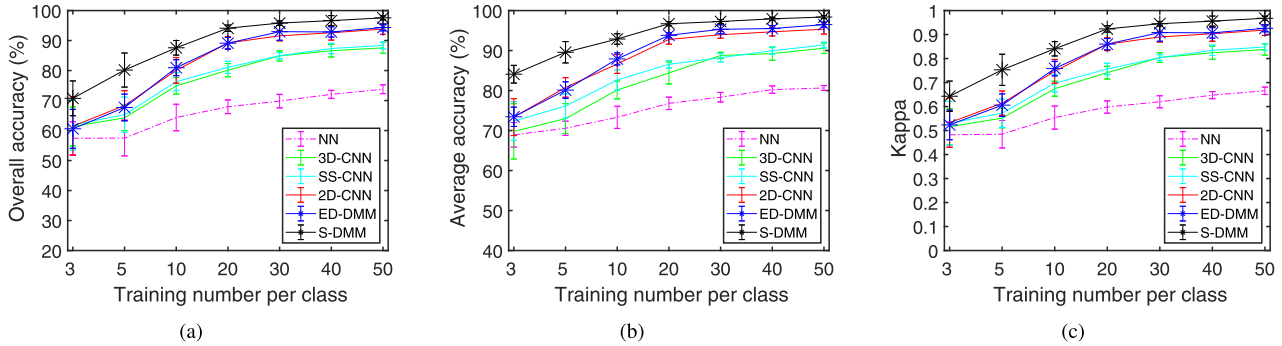


Fig. 7. PaviaU data set. The classification performances under different numbers of training samples. (a) OA. (b) AA. (c) Kappa.

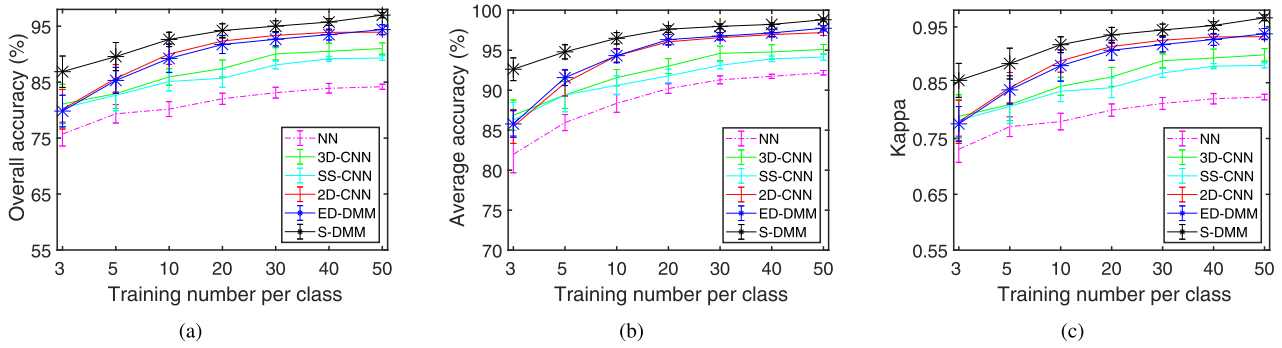


Fig. 8. Salinas data set. The classification performances under different numbers of training samples. (a) OA. (b) AA. (c) Kappa.

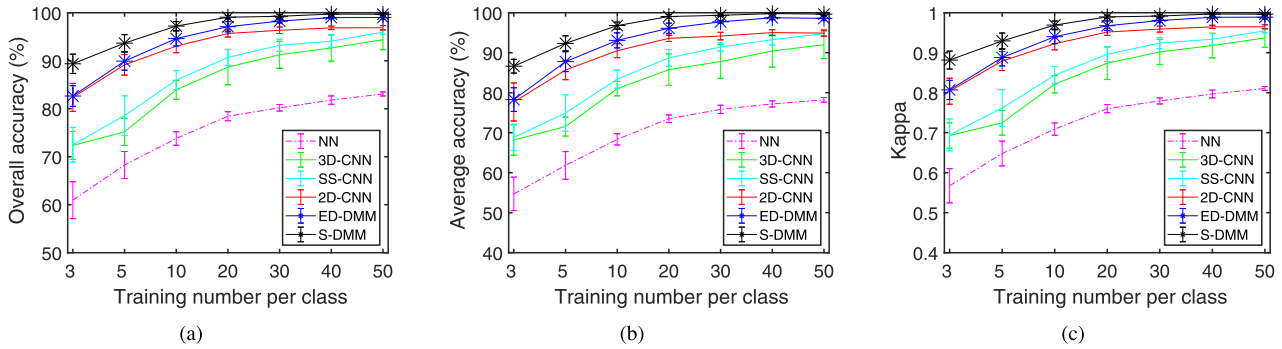


Fig. 9. KSC data set. The classification performances under different numbers of training samples. (a) OA. (b) AA. (c) Kappa.

for classification. However, by further learning the similarity between two samples rather than exploiting the Euclidean distance, the classification performance of S-DMM model shows a great improvement than that of ED-DMM, especially when the training sample is smaller. It is worth noting that although these advanced deep learning approaches, such as 3-D CNN [34] and SS-CNN [33], can achieve very good results as shown in their articles, and their training samples used for learning are large (e.g., more than 200 samples per class). The results of 2-D CNN are satisfactory to the same extent because it applies a data augmentation technique. However, our proposed S-DMM method can produce state-of-the-art results although the number of training samples is small and does not require any data augmentation. For example, when the number of training samples reaches to 50 per class, the classification accuracy of our S-DMM model can achieve very close to 100% for all three data sets, as shown in Figs. 7–9. More visually, Fig. 10 shows the classification maps of all these methods in a single experiment on the Pavia University data set. Obviously, by comparing Fig. 10(f) and (g), we can see that data augmentation plays a key role in the 2-D CNN method, helping it to make a great improvement in results. Even so, the classification map [Fig. 10(i)] of our S-DMM method is still the best accordance with the ground truth [Fig. 10(b)], achieving the overall accuracy of 98.89%. For brevity, the classification maps in other two data sets are omitted.

Tables V–VII illustrate the classification results of each class in detail when only ten samples per class are randomly selected from the labeled set for training. It is clear that the S-DMM method significantly outperforms others in all measures, including OA, AA, and κ. Furthermore, the

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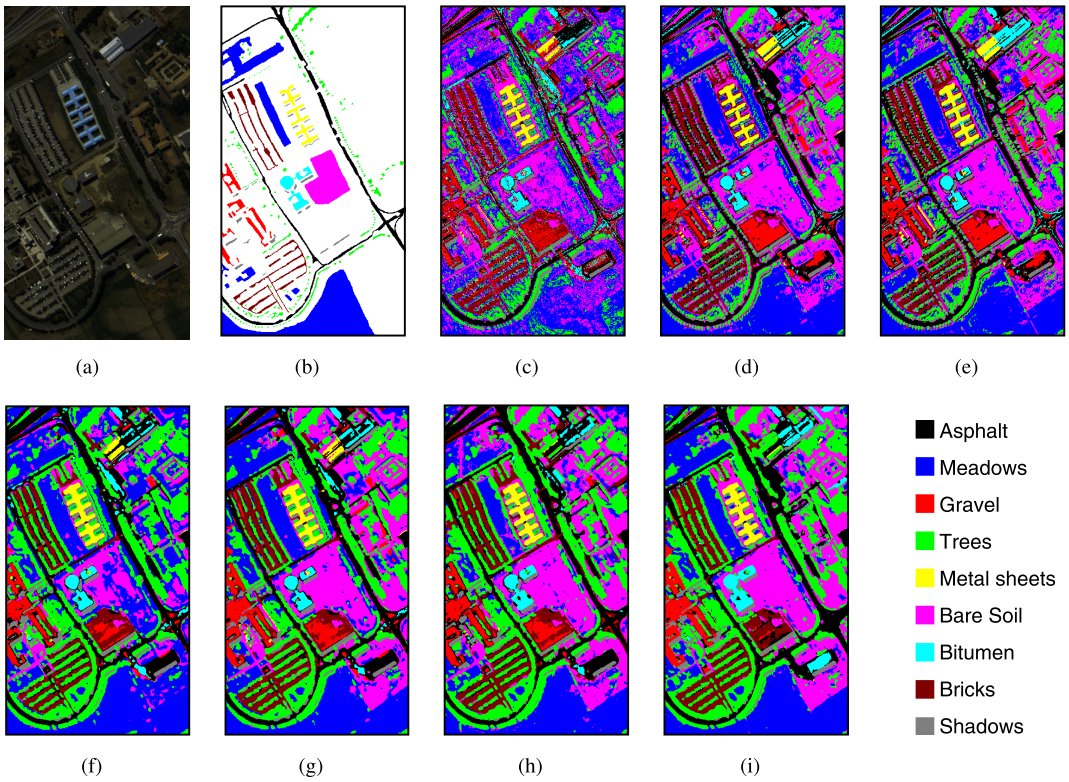


Fig. 10. Visual results. PaviaU data set. (a) False-color image. (b) Ground-truth map. Classification maps obtained by (c) NN (72.80%), (d) 3-D CNN [34] (91.35%), (e) SS-CNN [33] (90.63%), (f) 2-D CNN (without data augmentation) [37] (91.49%), (g) 2-D CNN [37] (96.99%), (h) ED-DMM (94.92%), and (i) S-DMM (98.89%).

TABLE IV
NAME AND SAMPLES' SIZE OF EACH LAND-COVER CLASS IN THE KSC DATA SET

Class	Name	Samples Size
C1	Scrub	761
C2	Willow swamp	243
C3	Cabbage palm hammock	256
C4	Cabbage palm/oak hammock	252
C5	Slash pine	161
C6	Oak/broadleaf hammock	229
C7	Hardwood swamp	105
C8	Graminoid marsh	431
C9	Spartine marsh	520
C10	Cattail marsh	404
C11	Salt marsh	419
C12	Mud flats	503
C13	Water	927
	Tatal	5211

classification accuracy of each class shows that the S-DMM method presents a higher precision and a lower deviation than others in most cases, indicating the effectiveness and robustness of the proposed method. We can also see that directly applying an NN classifier to spectral features shows the worst results on all data sets, but a great improvement can be seen when it applies to the embedding features (our ED-DMM method), which demonstrates the powerful feature mapping ability of our proposed embedding module. All these experimental results demonstrate the advantages of our metric learning-based models for the task of the same-scene classification.

B. Parameter and Convergence Analysis

In this section, a sensitivity analysis of two important parameters P (the patch size of the pixel) and D (the number of convolutional filters in the network) is performed. As shown in Fig. 11, the effect of the patch size and the number of filters on the performance of the proposed S-DMM method is given. For the patch size, we can see from Fig. 11(a) that the OA generally becomes stable as the increase in the patch size from 5 to 9. However, the OA of the proposed method reduces significantly when the patch size drops from 3 to 1. This is because only spectral information is used if the patch size is 1, which implies the importance of spatial information of each pixel. In terms of the number of convolutional filters, as shown in Fig. 11(b), we can clearly see that the OA is not much affected by the variation of the number of filters in a range between 32 and 128. We also present the convergence performance of the proposed S-DMM method in Fig. 11(b), which shows that our S-DMM approach converges smoothly in the training process.

C. Cross-Scene Classification

In this section, we will validate the proposed ED-DMM-UDA model on two cross-scene data sets, including Indiana scenes and Pavia scenes. In this experiment, both the labeled samples from the source scene and the unlabeled samples from the target scene are used for training and the evaluation will be measured in the target scene. The following three comparing methods are used

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TABLE V

CLASSIFICATION ACCURACY (%) USING NN, 3-D CNN [34], SS-CNN [33], 2-D CNN [37], AND OURS FOR THE PAVIAU DATA SET ON THE TEST SET WITH TEN LABELED SAMPLES PER CLASS AS THE TRAINING SET

	NN		3D-CNN [34]		SS-CNN [33]		2D-CNN [37]		ED-DMM		S-DMM	
Class	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	63.72	5.39	67.68	7.90	74.02	6.42	83.13	6.06	88.14	4.62	**91.92**	4.21
C2	60.40	10.50	77.93	9.63	73.50	7.48	73.84	9.64	73.32	7.36	**81.05**	5.43
C3	58.83	9.01	73.25	14.35	78.09	7.44	77.32	8.59	81.48	8.82	**88.13**	5.22
C4	80.66	11.71	84.23	9.33	87.39	6.02	90.45	4.49	90.43	6.40	**93.09**	5.63
C5	99.17	0.20	98.79	2.51	97.44	1.86	99.28	1.03	99.50	0.68	**100.00**	0.00
C6	55.88	11.65	46.16	16.82	62.77	15.68	76.25	11.13	78.77	7.87	**90.12**	7.78
C7	83.12	7.87	89.66	3.97	88.41	3.15	91.92	3.45	94.61	2.34	**98.73**	1.53
C8	58.12	5.64	84.88	9.03	85.47	6.78	88.01	4.82	85.08	7.84	**92.96**	3.93
C9	99.88	0.09	98.68	0.85	96.71	2.32	99.65	0.37	99.73	0.33	**99.98**	0.04
OA	64.34	4.42	74.89	2.78	76.29	2.98	80.06	4.25	80.95	2.67	**87.59**	2.43
AA	73.31	2.77	80.14	2.23	82.64	2.54	86.65	2.35	87.90	1.50	**92.89**	1.30
Kappa	0.55	0.05	0.67	0.03	0.70	0.04	0.75	0.05	0.76	0.03	**0.84**	0.03

TABLE VI

CLASSIFICATION ACCURACY (%) USING NN, 3-D CNN [34], SS-CNN [33], 2-D CNN [37], AND OURS FOR THE SALINAS DATA SET ON THE TEST SET WITH TEN LABELED SAMPLES PER CLASS AS THE TRAINING SET

	NN		3D-CNN [34]		SS-CNN [33]		2D-CNN [37]		ED-DMM		S-DMM	
Class	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	96.95	1.14	96.99	0.82	94.91	2.86	98.48	1.06	98.31	1.03	**99.23**	0.89
C2	95.74	2.50	99.75	0.35	99.45	0.57	99.78	0.12	98.71	1.05	**99.83**	0.18
C3	90.12	5.55	92.60	5.48	92.39	6.05	94.48	5.47	95.50	4.30	**97.92**	6.20
C4	98.24	2.50	97.71	1.29	95.46	4.16	98.40	1.53	98.76	1.14	**98.66**	2.12
C5	93.60	3.40	92.99	5.71	93.21	4.98	96.53	3.40	97.73	1.74	**98.74**	1.23
C6	98.05	1.25	98.54	0.81	98.10	1.85	**99.66**	0.38	98.87	0.85	99.56	1.31
C7	97.64	1.76	97.65	1.14	96.86	1.33	98.98	0.31	99.53	0.33	**99.94**	0.05
C8	53.49	6.68	70.21	9.71	71.68	8.84	75.71	9.01	71.04	10.51	**79.26**	4.81
C9	94.28	2.13	95.00	2.81	93.67	2.78	98.31	1.34	97.85	1.91	**99.12**	1.41
C10	78.73	5.62	84.54	6.95	82.38	5.53	87.06	7.79	88.04	5.75	**93.79**	4.32
C11	89.95	6.10	92.83	5.93	91.14	5.31	93.53	7.42	95.67	5.75	**98.92**	1.77
C12	95.96	5.03	98.09	2.18	98.05	1.23	99.05	0.78	99.30	0.84	**100.00**	0.00
C13	97.26	1.64	95.62	5.93	95.20	6.14	98.02	1.63	98.36	2.03	**99.36**	0.90
C14	89.97	3.73	93.50	3.41	93.72	2.30	96.80	1.64	96.51	2.91	**97.84**	2.49
C15	57.66	10.75	65.37	10.89	61.64	9.06	77.91	6.92	78.92	6.73	**83.93**	4.62
C16	85.95	5.90	93.61	4.16	91.75	3.10	95.98	3.32	95.93	3.39	**97.62**	2.20
OA	80.18	1.33	85.93	1.48	85.12	1.68	90.02	1.84	89.24	2.53	**92.63**	1.28
AA	88.35	1.12	91.56	1.02	90.60	1.17	94.29	0.79	94.31	0.88	**96.48**	0.72
Kappa	0.78	0.01	0.84	0.02	0.83	0.02	0.89	0.02	0.88	0.03	**0.92**	0.01

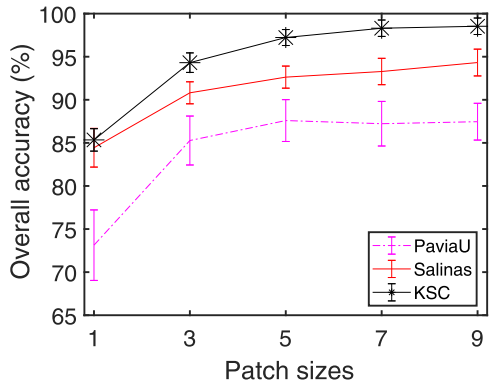
TABLE VII

CLASSIFICATION ACCURACY (%) USING NN, 3-D CNN [34], SS-CNN [33], 2-D CNN [37], AND OURS FOR THE KSC DATA SET ON THE TEST SET WITH TEN LABELED SAMPLES PER CLASS AS THE TRAINING SET

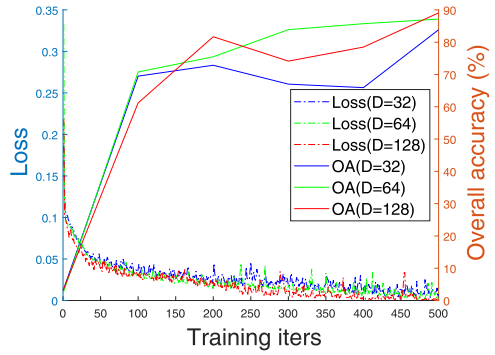
	NN		3D-CNN [34]		SS-CNN [33]		2D-CNN [37]		ED-DMM		S-DMM	
Class	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std	Mean	Std
C1	69.77	6.90	87.86	9.23	91.92	4.31	94.49	2.82	93.62	5.13	**94.79**	7.28
C2	67.42	13.84	83.86	2.93	84.98	3.39	89.79	4.68	91.29	5.31	**95.45**	5.26
C3	69.51	5.58	88.82	6.87	88.37	11.34	92.80	5.91	90.57	4.85	**94.27**	6.29
C4	41.57	5.14	45.29	10.51	55.66	8.80	71.86	16.54	77.98	10.49	**95.99**	2.51
C5	50.66	4.85	67.35	8.49	69.60	12.16	75.10	7.76	85.56	7.61	**93.05**	5.43
C6	40.73	5.56	62.33	18.87	69.54	8.37	84.34	6.48	93.11	5.77	**97.81**	2.66
C7	78.53	7.85	97.37	1.14	96.63	1.84	95.37	7.31	97.37	3.41	**97.47**	3.98
C8	59.31	7.96	82.78	8.84	86.72	5.89	94.20	4.34	**97.53**	2.59	96.98	3.30
C9	77.51	7.19	89.94	6.27	90.53	6.32	92.55	7.02	93.86	6.21	**97.43**	5.18
C10	73.27	5.16	82.03	7.26	85.41	6.01	96.52	2.65	97.11	3.09	**99.75**	0.54
C11	90.90	3.92	97.41	2.54	96.33	4.16	97.78	3.02	97.43	3.07	**99.41**	1.86
C12	71.20	5.98	75.66	11.42	76.61	9.11	92.01	2.06	94.77	5.32	**95.84**	3.29
C13	98.14	0.59	92.58	5.81	92.02	5.80	99.83	0.29	**100.00**	0.00	**100.00**	0.00
OA	73.79	1.41	83.94	1.97	85.98	1.97	93.04	1.37	94.62	1.55	**97.23**	0.93
AA	68.35	1.41	81.02	1.80	83.41	2.27	90.51	1.75	93.09	1.88	**96.79**	0.87
Kappa	0.71	0.02	0.82	0.02	0.84	0.02	0.92	0.02	0.94	0.02	**0.97**	0.01

1) *ED-DMM:* It is our proposed model for the same-scene classification as described earlier. Actually, the ED-DMM model is the same as the ED-DMM-UDA model but without applying unsupervised domain adaptation. It is trained on the training samples in source images and applied directly to the test samples

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(a)



(b)

Fig. 11. Parameters analysis of P (patch size) and D (the number of convolutional filters). (a) Performance of different patch sizes in three HSI data sets. (b) Convergence performances using different numbers of convolutional filters in the PaviaU data set. Both the results are achieved with ten training samples per class for training.

in the target images for prediction. Since the ED-DMM method does not use any domain adaptation, we regard it as the baseline.

2) *TSVM:* Transductive SVM (TSVM) is a conventional classifier level-based domain adaptation algorithm, which is achieved based on SVM by introducing transductive learning. In the cross-scene HSI classification, TSVM is first trained only by using the labeled samples in the source images and then, the unlabeled samples that are assigned with labels by the pretrained TSVM are added into the training set to retrain the classifier. In this way for several steps, the labels of unlabeled samples are reassigned and the TSVM can also be retraining during each iteration.
3) *SD-MTJDL-SLR:* It is a feature level-based domain adaptation method proposed in [43]. The goal of SD-MTJDL-SLR is to train a shared dictionary by samples in both source and target images by using its multitask joint dictionary learning model and then to train a sparse logistic regression (SLR) classifier based on the extracted features.

For SD-MTJDL-SLR and TSVM, we compare directly with the results of the comparative experiments presented in the article [43] to eliminate the influence on parameter settings.

The first experimental data are the Indiana scene, which were acquired from the AVIRIS sensor over the Northwest Tippecanoe Country, Indiana [63]. As the same as the article [43], two separate subsets are selected as the source and target scene images (both the scenes have the same size of 400 × 300 × 220 with 220 numbers of bands). Seven land-cover classes are shared by both scenes for classification, and the detailed information is shown in Table VIII and their ground-truth maps are shown in Fig. 12.

TABLE VIII
NAME AND SAMPLES' SIZE OF EACH LAND-COVER CLASS IN THE INDIANA SCENES

Class		Labeled samples	
	Name	Source scene	Target scene
C1	Concrete/Asphalt	4867	2942
C2	Corn cleanTill	9822	6029
C3	Corn cleanTill EW	11414	7999
C4	Orchard	5106	1562
C5	Soybeans cleanTill	4731	4792
C6	Soybeans cleanTill EW	2996	1638
C7	Wheat	3223	10739

TABLE IX
CLASSIFICATION ACCURACY ON THE INDIANA TARGET SCENE WITH 180 TRAINING SAMPLES PER CLASS IN THE INDIANA SOURCE SCENE

Algorithm	OA (%)	AA (%)	κ
ED-DMM	39.46	35.43	0.29
TSVM	39.19	33.82	0.27
SD-MTJDL-SLR	51.34	43.51	0.38
ED-DMM-UDA	**56.78**	**51.68**	**0.46**

TABLE X
NAME AND SAMPLES' SIZE OF EACH LAND-COVER CLASS IN THE PAVIA SCENES

Class		Labeled samples	
	Name	Source scene	Target scene
C1	Trees	266	2424
C2	Asphalt	266	1704
C3	Parking lot	265	287
C4	Bitumen	206	685
C5	Meadow	273	1251
C6	Soil	213	1475

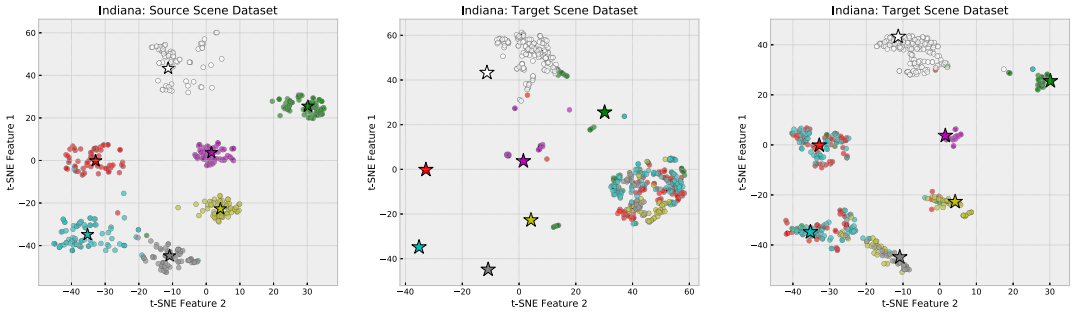
In the experiments, 180 numbers of source domain labeled pixels per class are selected randomly as a training set, and all the labeled pixels in the target scene image (testing set) are used to evaluate the classification performance. All the experiments are conducted in the same way as in the article [43], and Table IX lists the accuracies of different algorithms. The figures in Table IX demonstrate that the proposed ED-DMM model can achieve comparative results as the TSVM method, and ED-DMM-UDA can further improve the classification accuracy after applying unsupervised domain adaptation. Fig. 13(a)–(c) shows the clusters formed by the embedding features produced by our model from the Indiana source data, target data before domain adaptation, and target data after domain adaptation, respectively. In Fig. 13(a)–(c), we use the T-distributed stochastic neighbor embedding (t-SNE) [64] technique to visualize the embedding features in a 2-D space and several star shape points represent the cluster centers of source embedding features corresponding to different labels. Comparing Fig. 13(b) and (c),

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(a) (b) (c) (d)

Fig. 12. Source and target images of the Indiana scenes. (a) False-color image of source scene. (b) Ground-truth map of source scene. (c) False-color image of target scene. (d) Ground-truth map of target scene.



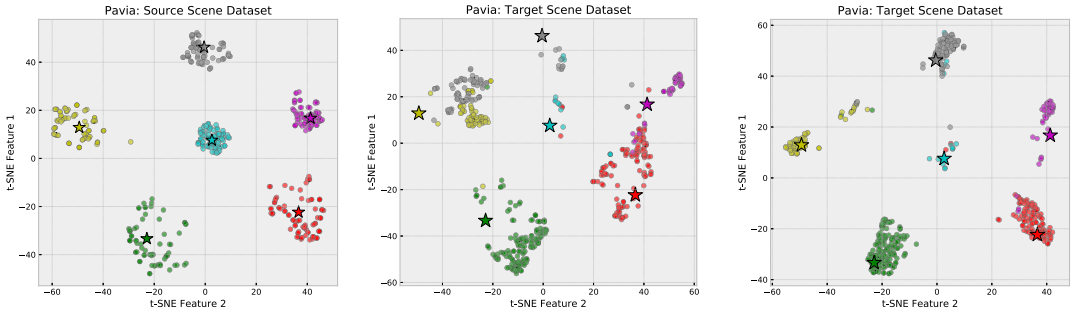
(a) (b) (c)

Fig. 13. Visualization of the t-SNE [64] components of the embedding features in the Indiana scenes data set. (a) t-SNE visualization of source embeddings. (b) t-SNE visualization of target embeddings before domain adaptation. (c) t-SNE visualization of target embeddings after domain adaptation. Colors represent different labels and stars represent the cluster centers of the source embeddings.



(a) (b) (c) (d)

Fig. 14. Source and target images of the Pavia scenes. (a) False-color image of source scene. (b) Ground-truth map of source scene. (c) False-color image of target scene. (d) Ground-truth map of target scene.



(a) (b) (c)

Fig. 15. Visualization of the t-SNE [64] components of the embedding features in the Pavia scenes data set. (a) t-SNE visualization of source embeddings. (b) t-SNE visualization of target embeddings before domain adaptation. (c) t-SNE visualization of target embeddings after domain adaptation. Colors represent different labels and stars represent the cluster centers of the source embeddings.

we can clearly see that the impact of distribution shift can be significantly reduced and the target embeddings [see Fig. 13(c)] can form similar clusters as the source embeddings [see Fig. 13(a)] after domain adaptation.

The second real HSI data set that we use is the Pavia scenes, which was captured by the digital airborne imaging system (DAIS) sensor during a flight over the urban areas of Pavia City, Italy. In the experiments, we select the small subset of

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TABLE XI

CLASSIFICATION ACCURACY ON THE PAVIA TARGET SCENE WITH 180 TRAINING SAMPLES PER CLASS IN THE PAVIA SOURCE SCENE

Algorithm	OA (%)	AA (%)	κ
ED-DMM	77.04	72.90	0.71
TSVM	61.21	61.50	0.53
SD-MTJDL-SLR	83.52	81.30	0.79
ED-DMM-UDA	**90.34**	**87.87**	**0.88**

Pavia University image (with the size of $243 \times 243 \times 72$) as the source scene image for training, and the Pavia Center image (with a larger size of $400 \times 400 \times 72$) is selected as the target scene image for prediction, as described in [43]. The land-cover classes of both scenes are listed in Table X and their false-colour images and ground-truth maps are described in Fig. 14. Following the same experimental settings as that on the Indiana data, the classification results of different approaches are shown in Table X. From the results on the Pavia data, we almost can get the same conclusion as described as on Indiana data. Another conclusion is that we can get better cluster embeddings on the Pavia target data, as shown in Fig. 15(c). All these experiments demonstrate that the proposed ED-DMM-UDA is powerful for the cross-scene HSI classification.

V. CONCLUSION

In the HSI classification, most of the deep learning-based methods require a large number of labeled pixels to train the classifiers. In this article, a well-designed network architecture based on metric learning is presented, which can tackle the problems of limited training samples. In addition, we propose a training strategy for learning the model in a simple way. Moreover, by applying an unsupervised domain adaptation technique, our proposed model can also handle the cross-scene HSI classification when the labeled pixels in target images are not existent. The proposed network model is tested on a series of popular HSIs and can achieve the outstanding results in both the same- and cross-scene classifications. Experimental results demonstrate that our proposed network is robust and a small number of hyperparameters can be fixed and are suitable for all kinds of data. In the future work, we will like to train our supervised model in an unsupervised way, since the learning process of the proposed model does not necessarily require exact labels of a pair of samples, as long as we know if they belong to the same class.

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